Performance Analysis of Company Teams and Employee Reassignment Using Sentiment Analysis

Problem statement

Many organisations do not have a proper or unified system to support HR and management in making decisions related to the replacement of employees or swapping them across teams. Presently, these decisions are mostly made based on managers' personal opinions. Due to this bias, the productivity of teams sometimes becomes lower.

The core problem is the lack of a data-driven decision support system that can evaluate employee skills, experience, performance, and satisfaction. By adding sentiment analysis and a structured feedback procedure, the system can assist HR in assigning suitable employees to suitable teams. The aim of this system is to create balanced teams and enhance the overall productivity of the company.

Introduction

In today's organizations, it is very common for different teams to work on similar projects, yet their performance levels are quite different. Some teams deliver work faster and more efficiently, while others struggle, even though they may have similar technical resources. These differences usually arise not only because of technical skills, but also due to factors such as employee experience, emotion, motivation, job satisfaction, and how well a person's skills fit within a team. Because of this, HR departments often face difficulty in making fair decisions related to employee role change or replacement. Most organizations evaluate performance based on numbers like meeting deadlines, work output, and manager opinions. Although these factors are important, they do not fully reflect the real working condition of employees. How employees feel about their work, their stress levels, morale, and comfort within a team often play a major role in productivity.

This project aims to address this issue by introducing a framework that combines sentiment analysis with traditional performance evaluation. By analysing employee feedback along with skills, experience, and performance data, the system provides a more realistic understanding of employee behaviour and team dynamics. This helps organizations to take balanced decisions, which will lead in better team performance and a healthier working environment.

Proposed Methodology –

The system is designed to evaluate the team performance and support HR in decision making i.e reassignment and swapping of employee between teams.

The system is divided into three main phase –

Phase-1 Employee Data Collection and Preprocessing –

-> Employee-related data is collected from sources

- Structured data : employee skills, years of experience, and past performance.

- Unstructured data : feedback, responses to surveys, social media posts, etc.

-> Textual feedback is cleaned by removal of noise , tokenization and normalization.

-> Structured data is standardized for consistency.

Phase-2 Sentiment Analysis (BERT (cls)) –

-> **Natural Language Processing techniques** are used to analyze the feedback to determine the satisfaction of employees towards their work.

-> This will produce results as positive, negative, and neutral (fuzzified)

-> This score will reflect the employee's emotional state and overall satisfaction towards the current team.

Phase-3 Performance Evaluation and Reassignment –

-> Profiles of employees will be created based on skill set, satisfaction score, and experience.

-> These profiles are analyzed at the team level to identify problems like skill mismatch, low morale, etc**.**

Final Recommendation –

-> The system will generate recommendations for reassignment or swapping of employees between teams.

-> If needed, it will recommend termination also.

Workflow Diagram

